

Sefa Demirtürk

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PROFILE

Electronics Engineer with experience in FPGA design, embedded systems, digital signal processing, cybersecurity, artificial intelligence and network technologies. Experienced in SIEM platforms, threat analysis, machine learning applications and defense industry projects. Strong background in VHDL/Verilog development, DSP algorithms and real-time system implementation.

EXPERIENCE

Koç Information and Defense Technologies

Oct 2024 – Sep 2025

System Management

- FPGA development using Xilinx FPGA platforms.
- Designed UART and I2C communication interfaces.
- Developed DSP applications including FIR/IIR filters and FFT.
- Performed MATLAB simulations and fixed-point verification.
- Worked on sonar signal processing applications.

KoçSistem (Koç Holding)

Nov 2023 – Oct 2024

SOC Project Assistant

- Collected, parsed and normalized security logs using IBM QRadar SIEM.
- Developed correlation rules and monitored security offenses.
- Integrated firewall, IDS/IPS and endpoint security logs.
- Conducted threat intelligence integration and alarm tuning activities.

Turkcell

Sep 2022 – Oct 2023

Network Department – Candidate Engineer

- Performed network performance monitoring using SNMP-based tools.
- Implemented traffic prioritization and bandwidth management using Cisco QoS.
- Conducted application traffic analysis and load balancing using Citrix ADC.
- Assisted in root cause analysis for network performance issues.

TÜBİTAK SAGE

Feb 2021 – Sep 2022

Electronics Manufacturing and Artificial Intelligence Department

- Developed UART and I2C communication modules using VHDL on FPGA platforms.
- Built machine learning applications using Python.
- Performed data cleaning and feature engineering using Pandas and NumPy.
- Developed deep learning models using TensorFlow and Keras.

PROJECTS

Autonomous Recycling System with Image Processing

- Developed an object-recognition based waste sorting system using Raspberry Pi.
- Implemented image processing and object classification using OpenCV and YOLOv5.
- Trained machine learning models with PyTorch.
- Integrated embedded hardware with AI inference systems.

Real-Time Network Traffic Anomaly Detection System

- Developed an AI-based intrusion detection system.
- Used NSL-KDD and CICIDS2017 datasets.
- Achieved F1-score of 0.96 using XGBoost.

- Built a real-time architecture using Flask and Kafka.

Autonomous Marine Cleaning Vehicle – Teknofest 2021

- Designed autonomous navigation using GPS and compass modules.
- Developed wireless communication infrastructure.
- Implemented route planning and energy optimization methods.

SKILLS

Programming

Python, C, C++, VHDL, Verilog HDL, Bash

FPGA & Embedded Systems

Xilinx FPGA, Nexys A7, Raspberry Pi, UART, I2C, GNSS

Digital Signal Processing

FIR Filters, IIR Filters, FFT, DFT, MATLAB

Artificial Intelligence

PyTorch, TensorFlow, Keras, Scikit-Learn, YOLOv5

Cyber Security

IBM QRadar SIEM, IDS/IPS, Threat Intelligence

Data Engineering

Pandas, NumPy, Apache Spark, Kafka, Grafana, Prometheus

Tools

Docker, OpenCV, Flask, Jupyter Notebook, Google Colab

EDUCATION

Gebze Technical University

2019 – 2026

B.Sc. in Electronics Engineering

CERTIFICATIONS

- Lifelong Cyber Security Expertise Certificate – Boğaziçi University BUSIBER
- Cyber Take-Off Governance Risk Compliance – THY Technology
- Supervised Machine Learning: Regression and Classification – Coursera
- Acoustics Training – Koç Information and Defense
- Quantum Horizons: Developments and Opportunities – TÜBİTAK

LANGUAGES

Turkish (Native)

English (Intermediate)